

Yongye(Felix) Hu

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EDUCATION

University of California, Berkeley

Master of Engineering | Robotics Engineering

Aug 2024 - May 2025

Berkeley, CA

Wuhan University of Science and Technology

Bachelor of Engineering (Honours) | Mechatronics Engineering

Sep 2020 - May 2024

Wuhan, China

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform

Sep 2024 - May 2025

Robotics Engineer & Project Lead

Berkeley, CA

- Led a 5-member cross-functional team to develop a hybrid flying-driving robot platform, delivering a fully functional prototype within 10 weeks.
- Developed a ROS2-based robotic system integrating a simulator and MPC controller for precise obstacle avoidance and full trajectory completion with real-time state logging.
- Designed and implemented a unified drivetrain supporting both aerial and ground locomotion using the same motor-gear system, reducing mechanical complexity and cost.

ROBOCON: Multi-Robot Coordination System

Sep 2022 - Jul 2023

Lead Mechanical Engineer & Embedded Developer

Wuhan, China

- Managed three interdisciplinary teams over a six-month project, overseeing a \$20,000 budget; delivered two autonomous ring-handling systems and one joystick-controlled quadruped within all milestones.
- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.
- Modularized the robot into gimbal, power, launcher, and chassis subsystems to achieve independent functionality and decoupled operation, incorporating modular interfaces and redundancy to improve system stability and debuggability.

Vacuum Dryer System: Concept to Prototype

Mar 2023 - May 2024

Lead Mechanical Engineer

Wuhan, China

- Spearheaded mechanical design and development of an innovative vacuum dryer system from concept to working prototype, enabling 20-minute drying at 40 °C with 50 dBA noise level.
- Developed a plug-and-play sensor-control module with real-time diagnostics, enabling seamless integration into the system and accelerating hardware/software debugging cycles.
- Engineered a magnetically coupled drive system that eliminated direct mechanical interfaces, resolving vacuum leakage at drive connections and ensuring torque-stable roller actuation.

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd.

Sep 2023 - Mar 2024

Mechatronics Engineer Intern

Wuhan, China

- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Integrated sensors including IR, microswitches, and pressure transducers to monitor system state and trigger alarms, enhancing operational reliability and safety.

SKILLS

Controls & Robotics: MPC, PID/Fuzzy PID, Trajectory Optimization, Closed-Loop Feedback Control, Sensor Fusion, ROS2, Inverse Kinematics

Embedded & Real-Time Systems: STM32, ESP32, Embedded C/C++, PCB Design (KiCad), CAN, UART, DMA, Real-Time Control

Programming: Python, C++, Linux, MATLAB, Qt

HW/SW Co-Design: SolidWorks, Autodesk Fusion, Ansys FEA/CFD, FDM/SLA 3D Printing, CNC Machining, DFA/DFM